Contour-Guided Hybrid Graph Network for Building Instance Segmentation From Airborne LiDAR Point Clouds

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***Abstract*—Driven by the growing demand for high-precision 3-D spatial data in smart cities, accurate building instance segmentation from airborne light detection and ranging (LiDAR) has become critical. Traditional methods often struggle in complex urban environments, while existing deep learning approaches primarily rely on local geometry and neglect explicit building morphology. To address these limitations, we propose a contour-guided hybrid graph network (CG-HGN) for building instance segmentation. CG-HGN incorporates contour to explicitly encode building morphology and leverages a hybrid graph architecture: a superpoint graph captures semantic features, and an edge graph models intersuperpoint boundary connectivity. By formulating instance segmentation as a graph connectivity inference problem, CG-HGN improves boundary precision and alleviates over-segmentation. Experiments on four datasets with distinct characteristics demonstrate that CG-HGN consistently outperforms existing methods in terms of average precision (AP) and mean coverage (mCov), particularly in sparse, heavily occluded, or structurally complex areas. Cross-dataset evaluations further confirm the strong generalization ability. CG-HGN provides an effective solution for high-quality building instance segmentation in real-world airborne LiDAR applications.**

***Index Terms*—Building instance segmentation, contour features, graph network, point cloud.**

I. Introduction

With the rapid advancement of urban digitization and smart city development, the demand for high-precision 3-D spatial data continues to rise [1], [2]. Light detection and ranging (LiDAR) enables fast and accurate acquisition of 3-D geometry and reflectance information, making it highly suitable for urban modeling [3], object recognition [4], and spatial information extraction [5]—particularly for building mapping and instance segmentation in complex urban environments. As fundamental units of urban space, buildings play essential roles in applications such as low-altitude flight-path planning [6], [7], digital-twin city construction [8], [9], and population estimation [10]. Therefore, accurately and robustly extracting building instances from airborne LiDAR point clouds is of substantial scientific and practical importance.

Existing approaches for airborne LiDAR building instance segmentation fall into traditional methods and deep learning methods. Traditional pipelines typically follow a two-stage paradigm of building detection followed by single-object extraction. Detection often relies on geometric model fitting [11], [12], morphological filtering [13], [14], region growing [15], [16], or clustering [17], [18] to isolate candidate building regions. Instance separation is commonly performed by aggregating neighboring points based on manually defined distance, shape, or density thresholds [19], [20], [21], [22]. Among these methods, contour analysis cluster (CA-Cluster) [23], [24], [25], [26] offers a particularly perspective: by representing height isovalues, contour lines provide a unified description of planar and curved surfaces, enabling a holistic understanding of building massing and hierarchical structure even when partial façade information is missing. However, such methods remain sensitive to data quality and parameter settings, and their reliance on numerous manually tuned thresholds limits robustness under severe occlusions, uneven sampling, or complex urban morphology.

Deep learning has recently emerged as the dominant paradigm for 3-D instance segmentation, integrating semantic prediction (building detection) and instance partitioning (single-object extraction) into a single end-to-end framework. Although most existing models are designed for general 3-D instance segmentation, many existing models can still be adapted to building instance segmentation on airborne LiDAR data. Proposal-based methods predict instances in a coarse-to-fine manner [27], [28], [29], but the high degrees of freedom in 3-D bounding boxes often make poor proposals a performance bottleneck. Grouping-based methods adopt bottom-up strategies, such as offset prediction toward instance centers

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[30], [31], [32], [33], but commonly struggle with boundary precision. Transformer-based approaches represent instances via learnable queries [34], [35]; however, they typically require a large number of queries, which imposes substantial GPU memory cost. To mitigate the computational and memory burden at the point level, recent studies [36], [37], [38], [39] introduce superpoints (or learnable supertokens) as compact geometric units to replace raw point representations. Building on this idea, graph-based methods [40], [41], [42], [43] further construct superpoint graphs and perform message passing combined with global optimization strategies, ultimately achieving superpoint grouping based on node feature similarity. Despite these advances, existing deep models still rely heavily on local geometric or neighborhood cues, treating the graph primarily as a message propagation structure while overlooking the explicit geometric and boundary information embedded within it. Moreover, they lack mechanisms to directly encode the global morphological structure of buildings, making them prone to over-segmentation or under-segmentation in regions with similar shapes or tightly connected architectural components.

To bridge the gap between local geometric learning and global structural reasoning, we revisit the utility of height-isovalue contours. These contours are generated via a bottom-up slicing of the point cloud along the elevation axis, providing a discretized representation of building massing. Due to the discrete and sparse nature of airborne LiDAR, these raw contours inevitably exhibit local irregularities and jaggedness. Despite these local geometric imperfections, they robustly preserve the fundamental topological structure of building instances. By leveraging the powerful feature extraction capabilities of deep learning, we can effectively capture the hierarchical relationships encoded within these priors. This integration not only avoids the rigid hand-crafted thresholds of CA-Cluster but also provides explicit structural constraints that are lacking in existing graph-based representations.

Building on this insight, we propose the contour-guided hybrid graph network (CG-HGN). To fully exploit the structural cues provided by contours, our framework consists of a superpoint graph for semantic and geometric representation and an edge graph for explicit boundary modeling, where cross-graph feature interaction enables joint reasoning over semantic and boundary information. Unlike previous grouping-based methods that often compromise boundary precision due to their reliance on implicit center-offset regression or feature similarity clustering, our method utilizes these contour priors to explicitly model boundaries within the edge graph. Consequently, CG-HGN reformulates instance segmentation as a direct graph connectivity inference task. By predicting superpoint connectivity based on explicit boundary probabilities, the network avoids complex clustering or mask regression while effectively mitigating over-segmentation. Experiments on multiple datasets demonstrate that CG-HGN achieves superior accuracy and robustness, with strong generalization across diverse urban environments.

The main contributions of this work are as follows.

1) *Integration of Contour Priors via a Dynamic Gating Mechanism:* We introduce contours as explicit morphological priors and adaptively fuse them with local geometric features. This ensures robustness against sparsity and occlusion in complex urban environments.
2) *Design of the CG-HGN:* This framework jointly models superpoint semantics and intersuperpoint boundaries. Through cross-graph feature interaction, it enhances precision and effectively reduces over- and under-segmentation errors.
3) *Reformulation of Building Instance Segmentation as a Graph Connectivity Inference Task:* We replace complex clustering algorithms with direct edge connectivity prediction. This approach eliminates cumbersome post-processing, enabling the efficient extraction of building instances.

II. Related Works

As traditional geometry- and rule-based methods for LiDAR building extraction have been extensively reviewed elsewhere [44], we focus here on deep learning-based 3-D instance segmentation approaches. These methods form the methodological foundation upon which our framework is built.

A. Proposal-Based Methods

Proposal-based 3-D instance segmentation typically follows a coarse-to-fine two-stage paradigm, where candidate regions are first detected and then refined. For example, 3D-SIS [45] extends 2-D bounding-box detectors to 3-D by predicting volumetric 3-D boxes from voxelized features, followed by point-level mask refinement within each box. 3DBoNet [29] employs Hungarian matching [46] to associate predicted axis-aligned bounding boxes (AABBs) with ground-truth (GT) instances and fuses AABB features with point features to generate instance masks. GSPN [47] selects query points via set abstraction and predicts an AABB for each query before performing point-level segmentation inside the box. Recent work such as SphericalMask [28] replaces traditional Euclidean bounding boxes with spherical polygon approximations, which reduces unnecessary access to background points. A common limitation of these approaches is their strong reliance on proposal quality. When the predicted box is misaligned with the true object boundary or suffers from localization errors, subsequent mask refinement tends to produce false negatives (FNs) or false positives (FPs), thereby constraining the overall performance ceiling.

B. Grouping-Based Methods

Grouping-based approaches learn point or superpoint embeddings and geometric offsets, and then form instances through clustering. PointGroup [30] first introduced centroid-offset prediction and performs clustering on the shifted point cloud. SoftGroup [31], [32] leverages soft semantic scores to guide grouping and further refines positives while suppressing FPs in a top-down refinement stage. PBNet [48] assigns points with different density levels: high-density points grow instance clusters, while low-density points vote for membership. DyCo3D [33] represents each instance with a dynamic convolution kernel that is adaptively generated for each cluster

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to predict instance masks. To mitigate error propagation during clustering, ISB-Net [38] adopts instance-aware farthest-point sampling (IA-FPS) to improve the robustness of dynamic kernels. To improve efficiency, B-Seg [49] exploits point-feature similarity and performs grouping via a relation matrix, avoiding the need to process all points in large-scale scenes. Although these methods avoid explicit proposal generation, their performance is often sensitive to embedding separability and clustering or threshold hyperparameters, making them prone to cluster drift when generalizing across diverse scenes.

C. Transformer-Based Methods

Inspired by recent progress in 2-D instance segmentation [50], [51], [52], 3-D Transformer-based approaches represent each object instance using a learnable query vector. Through iterative attention interactions in the decoder, these queries are progressively updated to produce instance masks. Early works such as Mask3D [53] and SPFormer [37] initialize queries using farthest-point sampling and learnable embeddings, respectively. Subsequent methods—including QueryFormer [34] and MaFT [54]—further refine query distribution and interaction strategies, while Relation3D [55] enhances the modeling of internal relationships between scene features and instance queries. These approaches provide strong global modeling capability; however, they typically require a large number of queries and substantial GPU memory, which leads to efficiency–accuracy tradeoffs when applied to large-scale urban scenes.

D. Graph-Based Methods

Graph neural networks (GNNs) have been widely applied to 3-D semantic segmentation, while their use in instance segmentation is comparatively less explored. DGCNN [56] introduces dynamic graph construction in feature space, and subsequent works enhance instance separability by incorporating channel–spatial attention mechanisms. More recently, ASGFormer [57] couples graph attention with dummy nodes for global propagation, improving separation of structurally adherent objects. Since point-level graph construction incurs high computational cost, recent efforts improve efficiency and robustness by first partitioning point clouds into geometrically homogeneous superpoints, and then performing learning on the superpoint graph. For example, superpoint Transformer (SPT) [41] builds superpoints and adjacency graphs based on geometric consistency for semantic prediction, upon which SuperCluster [42] predicts edge affinities and applies graph-cut to cluster superpoints into instances. EZ-SP [58] replaces CPU-bound cut-pursuit with a fully GPU-based learnable partition, markedly accelerating end-to-end inference. Other methods project 2-D segmentation results onto 3-D to form "pseudo-superpoints," followed by graph-cut-based instance extraction [59]. City-BIS [43] couples voxel–point fused semantic segmentation with a structure-aware graph clustering algorithm for robust instance separation in dense urban scenes. However, most existing graph-based approaches treat edges merely as adjacency and message-passing channels, thus under-utilizing the geometric and boundary semantics inherently carried by edges. Furthermore, relying solely on local geometry still leads to over- or under-segmentation when dealing with buildings of similar shapes or physically connected structures.

E. Integration With Segment Anything

The segment anything model (SAM) [60] and its variants provide powerful prompt-based segmentation capabilities in 2-D images, motivating a range of "SAM-assisted 3-D segmentation" paradigms. One line of work projects point clouds onto 2-D views, uses SAM-generated masks as priors or pseudo-labels, and back-projects them to 3-D to guide clustering or graph cut, as in [59]. Another direction leverages multiview images or image–point cloud fusion, using SAM masks to improve instance proposals or serve as auxiliary supervision [61], [62]. However, standard airborne LiDAR datasets (e.g., WHU3D provided by Han et al. [63]) typically lack the RGB information required by SAM, rendering such methods inapplicable. Moreover, projecting 3-D geometry to 2-D introduces severe occlusion and misalignment errors. Thus, given our focus on pure geometric segmentation, image-dependent SAM approaches fall outside the scope of this work and are not directly comparable.

III. Methods

The overall architecture of the proposed CG-HGN is illustrated in Fig. 1. First, we construct a hybrid graph representation by partitioning the point cloud into superpoints to build a superpoint graph and transforming its edges into nodes to form an edge graph. This step establishes the foundation for the joint modeling of regionwise semantics and boundary-level structures. Second, we integrate contour-guided morphological features to supplement local geometric descriptors. This allows for the adaptive fusion of global structural priors with local cues, ensuring robust feature representation. Third, we process these features within a hybrid framework, where the superpoint graph targets semantic prediction and the edge graph focuses on instance-boundary learning. Through cross-graph feature interaction, the two graphs mutually reinforce one another to refine feature discriminability. Finally, during inference, we derive building instances by performing connectivity analysis over the predicted boundary edges.

A. Superpoint Graph Construction

We adopt the SPT (proposed by Robert et al. [41]) as the backbone of the superpoint graph branch, including its superpoint partitioning, multiscale graph construction, and Transformer-based graph encoder. The input point cloud is first clustered into locally homogeneous superpoints using cut-pursuit [64], [65], yielding a superpoint graph $G_s = (\mathcal{V}_s, \mathcal{E}_s)$, where nodes correspond to superpoints and edges connect pairs of superpoints that contain spatially adjacent raw points.

The node features of G_s are initialized using the local geometric descriptors F_s^g defined in SPT. These features are subsequently fused with the contour-guided morphological descriptors through the fusion method described in Section III-B to obtain the final node representation.

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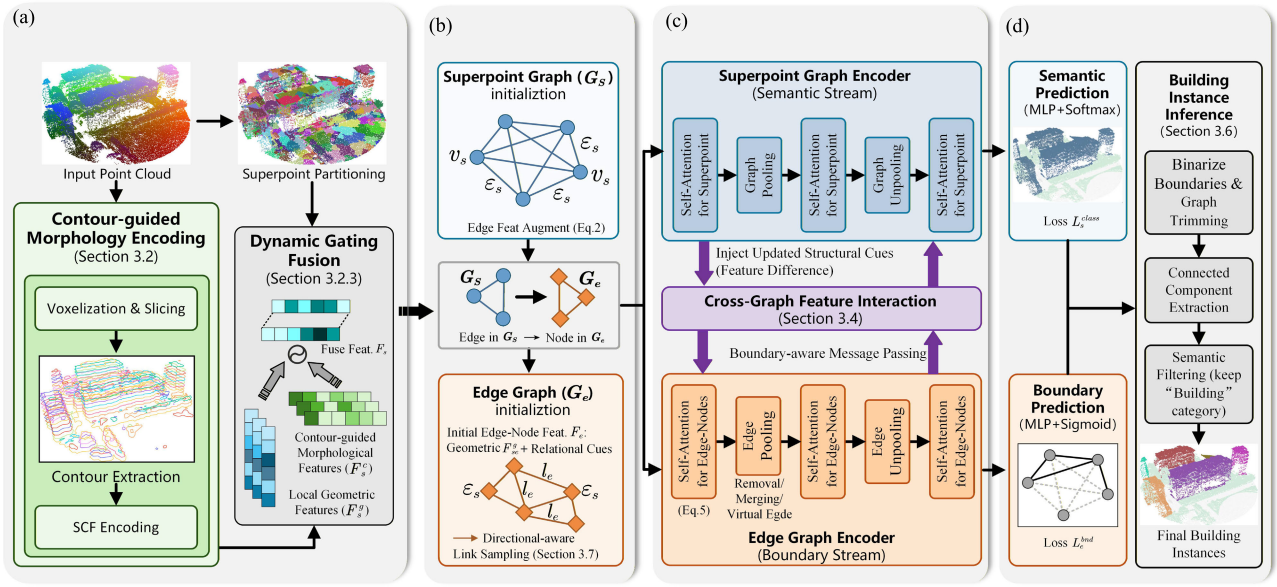

Fig. 1. Overall architecture of the proposed CG-HGN. (a) Data preprocessing and feature embedding. (b) Graph construction. (c) Hybrid graph reason framework. (d) Building instance inference.

By recursively partitioning superpoints, SPT forms a multiscale superpoint graph hierarchy that captures structures at different spatial resolutions. We adopt the same hierarchical structure and Transformer encoder as the backbone of our superpoint graph branch. Building upon this backbone, we further construct the corresponding multiscale edge graph and design a cross-graph interaction mechanism, which enhances the feature extraction capacity of the hybrid graph framework (see Section III-C).

B. Contour-Guided Morphology Feature Encoding

To enhance the model's ability to capture global building morphology, we introduce contour-guided morphological descriptors that characterize the outline and structural form of each superpoint region.

1) Contour Extraction: Traditional DEM-based contour extraction relies on rasterization and interpolation, which often introduce artifacts in areas with abrupt elevation changes or missing data (e.g., rooftop edges and skybridges). Addressing these issues, we employ a bottom-up slice-based approach. The point cloud is voxelized with a predefined voxel size and partitioned into horizontal slices along the z-axis. A local height field is constructed for each slice, and stable contours are extracted using open library GDAL [66]. Fig. 2 shows that this approach considerably alleviates interpolation artifacts while maintaining structural continuity.

2) Superpoint-Based Contour Family (SCF) Features: A superpoint S_i typically covers multiple adjacent contours, which we refer to collectively as an SCF. To construct the SCF, each raw point is first assigned to the contour polygon containing it within the same elevation slice via a point-in-polygon test. Since a superpoint comprises a cluster of raw points, its SCF is defined as the unique union of all contours corresponding to these constituent points. For each contour $c_j \in SCF(S_i)$, we compute geometric attributes (area A_j and centroid G_j) and shape attributes including rectangularity

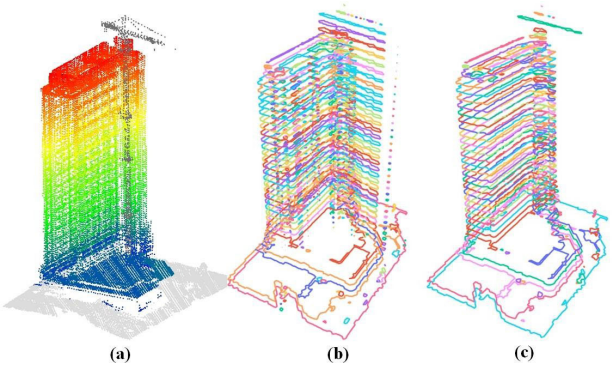

Fig. 2. Results of contour extraction. (a) Input point cloud. (b) Extraction based on DEM. (c) Extraction produced by our approach.

$R_i = (A_j/A_{mr})$, convexity $V_j = (A_j/A_{ch})$, and compactness $C_j = (L_j/\sqrt{A_j})$, where A_{mr} and A_{ch} denote the area of the minimum bounding rectangle and the convex hull, respectively, and L_j denotes the contour perimeter.

To obtain a unified-length shape representation, we compute the geometric union of all contours within an SCF on the XOY plane, obtaining $U_c = \bigcup_i c_i$, which is treated as the representative shape of the corresponding superpoint. A Fourier-based shape descriptor (FSD) [67] is then applied to encode U_c into a fixed-length vector Γ_s. This vector, together with the area A_s and centroid G_s of the representative contour, constitutes the geometric–morphological attributes of the superpoint. In addition, the shape attributes are aggregated using an area-weighted average, $\bar{f}_s = \left(\sum A_i f_i / \sum A_i\right)$, $f_i \in R_i, V_i, C_i$. Finally, the SCF descriptor is expressed as

$$F_s^c = \left[\Gamma_s, A_s, G_s, \bar{R}_s, \bar{V}_s, \bar{C}_s\right].$$

3) Feature Fusion: Each superpoint is represented by a combination of local geometric features F_s^g and SCF features F_s^c. However, these two feature types exhibit significantly different statistical distributions and gradient behaviors

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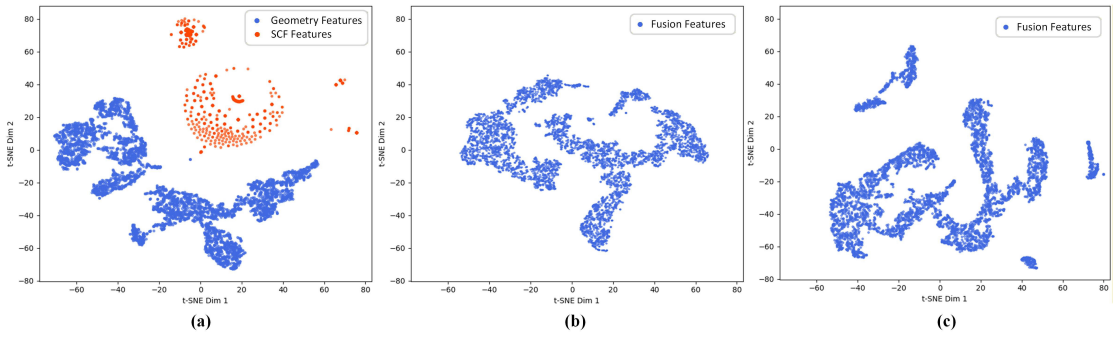

Fig. 3. Visualization of feature distributions before and after fusion. We employ t-SNE [68] to project high-dimensional feature vectors into a low-dimensional space to facilitate the visualization of feature evolution. (a) Input geometric and SCF features. (b) Dynamic gated fusion results, showing enhanced continuity. (c) MLP-based fusion results, exhibiting residual fragmentation.

(see Fig. 3). Geometric features tend to dominate early training due to their smooth and stable distribution, whereas SCF features are more irregular and difficult to optimize, leading to gradient competition. To mitigate this issue, we introduce a dynamic gating mechanism

$$F_s = \mathrm{MLP}\left(F_s^g, g_s \cdot F_s^c\right), \quad g_s = \sigma\left(W\left[F_s^g; F_s^c\right]\right) \tag{1}$$

where σ is the Sigmoid function and W is the learnable linear projection. This mechanism adaptively regulates the contribution of SCF features, enabling a smooth transition from local geometric perception to global morphological understanding.

For any pair of adjacent superpoints i and j, we further introduce contour-difference augmentation to the edge feature in the superpoint graph

$$F_{se} = \mathrm{MLP}\left(F_{se}^g,\ g_e \cdot \left(F_s^c(i) - F_s^c(j)\right)\right) \tag{2}$$

where g_e is the edge-level gating parameter. This enhances the ability of the model to explicitly capture morphological variation across adjacent regions.

C. Edge Graph Construction and Relational Learning

This section introduces the construction of an edge graph G_e and describes how edge–edge relational learning is achieved through the edge graph encoder, multiscale structural consistency, and an edge-level self-attention mechanism.

1) Edge Graph Design: At each scale, the edge graph constructs its nodes—referred to as edge nodes—directly from the edges of the corresponding superpoint graph. Formally, the edge graph is defined as $G_e = (\mathcal{E}_s, \mathcal{E}_e)$, where $\mathcal{E}_s$ denotes the set of edges in the superpoint graph. Two edges in $\mathcal{E}_s$ are considered adjacent in G_e if they share a common superpoint; accordingly, a link is established between their corresponding edge nodes in the edge graph (see Fig. 4). For clarity and to distinguish them from the topological elements of the superpoint graph, we explicitly refer to the nodes and edges within the edge graph as "edge nodes" and "links," respectively.

Edge-node features aggregate the local geometric edge descriptors F_{se}^g from SPT and the relational cues derived from its two incident superpoints. For an edge connecting superpoints S_i and S_j, two complementary components are computed from their features: 1) the absolute difference,

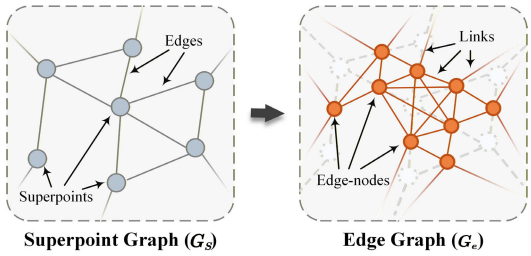

Fig. 4. Construction of the edge graph: superpoint graph edges are converted into edge nodes, and their links are determined by shared nodes.

which captures the magnitude of structural change and 2) the directional similarity, which reflects the similarity of feature directions. These components are concatenated with the geometric edge descriptor to form the enhanced edge-node feature

$$F_e = \mathrm{MLP}\left(F_{se}^g,\ F_{s_i} - F_{s_j},\ \cos(F_{s_i}, F_{s_j})\right). \tag{3}$$

Link features are constructed from four geometric attributes describing the relationship between adjacent edges: 1) length difference between edge direction vectors; 2) midpoint offset, defined as the spatial displacement between the midpoints of the two edges; 3) normal deviation, computed from the angle between their edge normals (with each edge normal approximated by the average normal of its incident superpoints); and 4) directional angle, measuring the angular deviation between the two edge direction vectors.

2) Edge Graph Pooling and Unpooling: The topology and feature of the edge graph depend on the multiscale transition of the superpoint graph. During the transition from fine to coarse scales, some fine-scale edges in the superpoint graph disappear as their incident superpoints are merged. Without correspondingly updating the topology and features of the edge graph, cross-scale feature extraction would become discontinuous, thereby disrupting the consistency required for subsequent attention operations and message passing. To address this issue, we design pooling and unpooling mechanisms for edge graph nodes, ensuring that the edge graph remains topologically and semantically aligned with the superpoint graph across scales. Fig. 5 visualizes how edges in the superpoint graph change with scale, which directly determines the creation, deletion, and merging of edge nodes.

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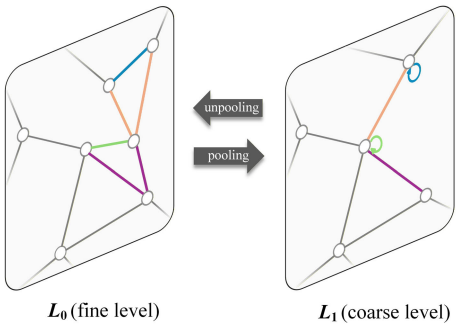

Fig. 5. Pooling and unpooling of edges in the superpoint graph. Colors denote the correspondence of edges across scales: blue/green edges are removed during pooling, whereas orange/purple edges are merged. Unpooling restores the fine-scale structure.

Edge pooling includes three cases.

1) *Edge Removal:* If two nodes are merged into a single parent node, the edge between them becomes invalid and is removed (corresponding to the blue/green examples).
2) *Edge Merging:* If nodes S_i and S_j are merged into new nodes S'_i and S'_j, all original edges connecting them are merged into a single edge (S'_i, S'_j) (orange/purple examples). The merged edge feature is aggregated via max pooling.
3) *Virtual Edge Creation:* To preserve local structural information, deleted edges are replaced with a self-loop edge e_{self}, whose feature is generated by average pooling.

Edge Unpooling: To recover fine-scale information, unpooling propagates coarse-scale features back to finer scales according to the recorded pooling mapping $A^{(l)}$. The upsampled feature is expressed as

$$F^{(l)}_{se\uparrow} = \text{MLP}\left(A^{(l)} F^{(l+1)}_e,\ F^{(l)}_e\right). \tag{4}$$

This operation maintains continuity across semantic levels and strengthens multiscale relational modeling in the edge graph.

3) Self-Attention Between Edge Nodes: Edge nodes inherently encode geometric structure. To explicitly learn these patterns, we transfer the self-attention mechanism used in the superpoint graph to the edge graph. For any adjacent edge-nodes pair $(e_i, e_j) \in \mathcal{E}_e$, we define three geometric bias vectors $A^{\text{key}}_{i,j}$, $A^{\text{query}}_{i,j}$, and $A^{\text{val}}_{i,j}$ for the key, query, and value embeddings. These biases encode directional differences, length variation, normal consistency, and angular deviation between edge nodes. The attention update for edge-node e_i is given by

$$[\mathcal{T}(m)]_i = \text{Attn}\Big(Q_i \oplus A^{\text{query}}_{i,\mathcal{N}(i)}, K_{\mathcal{N}(i)} + A^{\text{key}}_{i,\mathcal{N}(i)}, V_{\mathcal{N}(i)} + A^{\text{val}}_{i,\mathcal{N}(i)}\Big) \tag{5}$$

where $\mathcal{N}(i)$ is the adjacency set of e_i, and $\oplus$ denotes the broadcast addition. This formulation mirrors the attention mechanism in the superpoint graph, except that the normalization term is adapted from the number of neighboring superpoints to the number of neighboring edges—allowing the attention to naturally fit the structure of the edge graph.

D. Cross-Graph Feature Interaction

To jointly leverage superpoint-level semantics and edge-level geometry, we introduce bidirectional feature interaction between G_s and G_e after the graph encoding of each layer. The interaction process is described as follows.

1) Superpoint → Edge: After each layer updates the node features in the superpoint graph encoder, the feature differences between adjacent superpoints are recomputed and injected into the edge-node features. The operators used here—absolute feature contrast and directional similarity—are identical to those introduced in Section III-C1. Unlike the static differences used during initialization, these feature differences are computed on-the-fly during the forward pass, dynamically derived from the current-layer node features.

2) Edge → Superpoint: In the superpoint graph, edge features act as the carriers for message passing between superpoints. To make this propagation boundary-aware, the updated node features from the edge graph are used as boundary-aware factors and concatenated with the edge features of the superpoint graph. This enhancement provides the superpoint graph encoder with boundary-aware cues, which help reduce message passing across instance boundaries and encourage semantic consistency within each instance.

E. Loss Functions

CG-HGN is trained using two supervision signals.

1) Superpoint Semantic Classification: For superpoint s with label $c(s)$, the prediction is computed as $x_s = \text{softmax}(\text{MLP}(F_s))$, and the one-hot GT label is denoted by y_s. We optimize the standard cross-entropy loss

$$L^{\text{class}}_s = H(x_s, y_s). \tag{6}$$

2) Edge Graph Node Binary Classification: For each edge-node, the network predicts the probability of being an instance boundary as $b_e = \sigma(\text{MLP}(F_e))$, where $\sigma(\cdot)$ denotes the sigmoid function. The GT label is $\hat{b}_e \in \{0, 1\}$. Since boundary edges are significantly fewer than nonboundary edges, the samples are highly imbalanced. We adopt focal loss [69], which down-weights easy negatives and emphasizes hard samples

$$\begin{aligned} L^{\text{bnd}}_e = &-\alpha(1-b_e)^\gamma \hat{b}_e \log(b_e) \\ &-(1-\alpha) b_e^\gamma (1-\hat{b}_e) \log(1-b_e). \end{aligned} \tag{7}$$

In our experiments, we set $\alpha = 0.75$ and $\gamma = 2.0$.

The final loss is the weighted average of the semantic classification loss and the boundary classification loss

$$L = \frac{1}{|\mathcal{V}_s|} \sum_{s \in \mathcal{V}_s} L^{\text{class}}_s + \frac{1}{|\mathcal{E}_s|} \sum_{e \in \mathcal{E}_s} L^{\text{bnd}}_e. \tag{8}$$

F. Building Instance Inference

Unlike SuperCluster [42], which infers instance boundaries indirectly from superpointwise features, CG-HGN explicitly predicts boundary probabilities on the edge graph and uses them directly during inference. Specifically, for each edge nodes, the boundary probability produced by the edge graph encoder is binarized, and edges predicted as cross-instance

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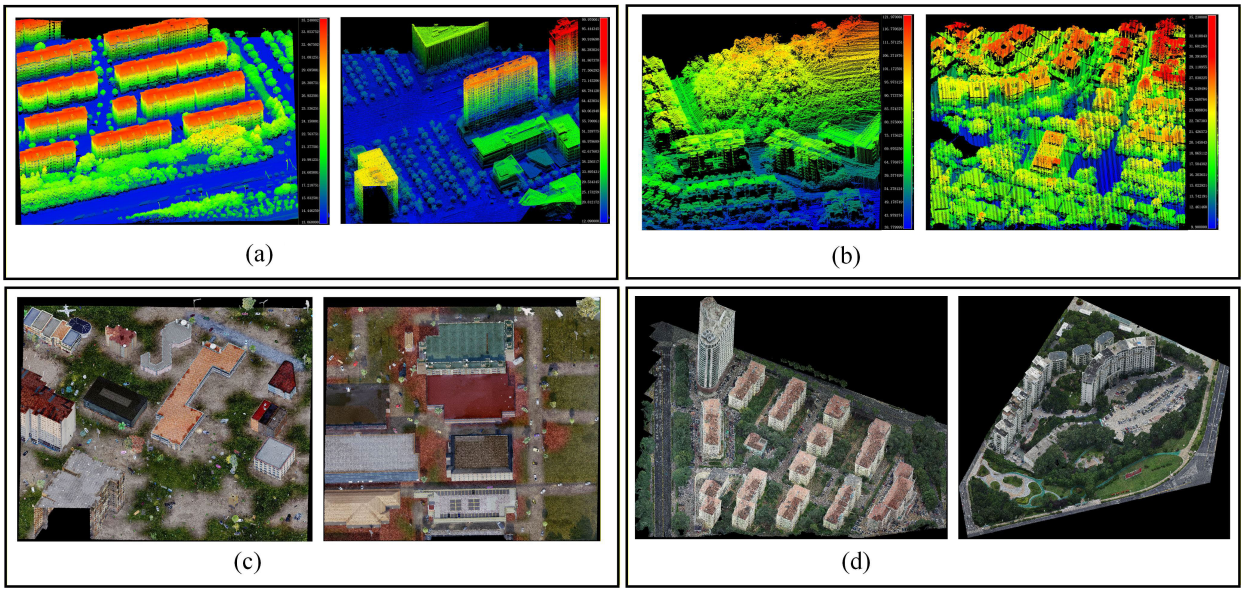

Fig. 6. Visualization of representative scenes from the experimental datasets. (a) WHU3D. (b) Guangzhou. (c) STPLS3D. (d) UrbanBIS.

boundaries are removed, leaving only intrainstance connections. Connected component extraction is then performed on the trimmed graph to obtain candidate instance regions. Since these regions may still include nonbuilding objects, superpoint semantic predictions are further applied to filter out nonbuilding components, and only those labeled as building are retained as final instance outputs.

G. Directional-Aware Edge Sampling

The number of edges in the edge graph G_e is usually much larger than the number of nodes in the point graph. Including all edges in training would significantly increase memory usage and computational cost. However, not all edges contribute equally to the model: edges located in planar regions usually exhibit high geometric consistency and contain redundant information, whereas edges in nonplanar regions encode more discriminative geometric variation. To preferentially retain edges with higher geometric complexity, we adopt a direction-aware sampling strategy based on normal variance.

For an edge e in the edge graph, its normal vector $\mathbf{n}_e$ is computed as the average of the normals of its two endpoint superpoints. Let $\mathcal{N}(e)$ be the set of its neighboring edges, and $\mathbf{n}_{e_k}$ be the normal vector of a neighboring edge e_k. We define the normal variance as

$$V(e) = 1 - \frac{1}{|\mathcal{N}(e)|} \sum_{e_k \in \mathcal{N}(e)} \left| n_e^\top n_{e_k} \right|. \tag{9}$$

A larger value of this measure indicates that the local region is less planar and exhibits more significant geometric variation. The sampling probability is then defined according to the normal variance as

$$P(e) = \frac{V(e)}{\sum_{e' \in \mathcal{E}_e} V(e')}. \tag{10}$$

In practice, we sample a fixed number of edges from the edge graph according to the probability $P(e)$. In this way, the model focuses more on training edges in geometrically complex boundary regions and reduces redundant edges in flat areas, thereby significantly decreasing memory consumption and training time.

IV. Experiments

A. Experimental Settings

1) Datasets: We evaluate the proposed CG-HGN on four urban datasets. According to the point-cloud acquisition modality, these datasets can be grouped into real world and synthetic data (see Fig. 6).

1) *WHU3D [63]:* This dataset is collected from real urban scenes in China. We use the airborne LiDAR point clouds with both instance and semantic annotations. Most building point clouds are well covered, while a small number of building façades exhibit missing points. The covered area is 2.7×10^6 m^2, and the point density is 60.82 pts/m^2.
2) *UrbanBIS [49]:* This dataset is generated by first reconstructing meshes from oblique photogrammetry and then sampling point clouds from the meshes. We use Qingdao region for training and testing. The covered area is 1.07×10^7 m^2, and the point density is 13.2 pts/m^2.
3) *STPLS3D [70]:* This is a synthetic dataset where virtual 3-D urban scenes are created in unreal engine, and point clouds are directly sampled from the scene models. Building point clouds are complete. The covered area is 6.0×10^6 m^2, and the point density is 13.2 pts/m^2.
4) *Guangzhou:* This dataset consists of real urban scenes collected by us in Guangzhou. Compared with WHU-Urban3D, a larger proportion of building façades points are missing. The covered area is 2.6×10^5 m^2, and the point density is 10.6 pts/m^2. We only manually annotate building instance labels.

Before training, all datasets are downsampled using a regular grid with a spatial resolution of 0.2 m. All reported metrics are computed on the original, nondownsampled point

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clouds. For the public datasets WHU-Urban3D, STPLS3D, and UrbanBIS, we strictly follow the official data splits. For Guangzhou dataset, we randomly split the data into training and testing sets with a ratio of 4:1.

2) Evaluation Metrics: For building instance segmentation, we adopt the standard average precision (AP) and mean coverage (mCov) metrics. Specifically, AP_{25}, AP_{50}, and AP_{75} denote the AP scores under IoU thresholds of 25%, 50%, and 75%, respectively. The mAP denotes the averaged scores with IoU thresholds from 50% to 95% with a step size of 5%. mCov (proposed by Silberman et al. [71]) measures the coverage of true-positive predicted instance masks with respect to the corresponding GT instance masks. In addition, we report semantic accuracy for building and nonbuilding categories. Here, mIoU-B is the IoU of the building class, while mIoU-S is the mean IoU over several nonbuilding (stuff) semantic classes. Since the nonbuilding semantic categories differ across the public datasets, to ensure the comparability and representativeness of the evaluation, we only consider four nonbuilding classes that are consistently available in all datasets: ground, vegetation, vehicle, and electrical infrastructure.

3) Model Parameterization: All experiments are conducted on a workstation equipped with a single 5090D GPU. We adopt a two-stage training scheme: the first stage (500 epochs) performs pretraining on the superpoint graph, and the second stage (1500 epochs) jointly trains the two graphs. The training batch size is set to 4, where each mini-batch consists of two randomly sampled spherical regions with a radius of 25 m. For contour extraction, the slice thickness is set to 1 m. For FSD, we retain the first ten coefficients as features. The superpoint graph uses the SPT64 [41] model configuration. During training, the number of sampled edges in the edge graph is set to twice the number of edges in the superpoint graph. We use AdamW [72] as the optimizer. The initial learning rate is set to 0.01, with a weight decay of 10^{-4}. A cosine warm-up is applied during the first 20 epochs, gradually increasing the learning rate from 10^{-4} to the initial value, followed by cosine annealing to decay the learning rate to 10^{-6}.

Both the proposed CG-HGN and SuperCluster [42] use the same superpoints and superpoint graph construction parameters as SPT64. Since none of the urban datasets provide superpoint labels, for the comparison methods ISBNet [38], SPFormer [37], Relation3D [55], and SphericalMask [28], we generate superpoints using the SPG [40] algorithm, with the query radius set to 10 m. For ISBNet [38], B-Seg [49], and SoftGroup [32], the clustering threshold is set to 5 m. CA-Cluster is reproduced following the implementation details and threshold parameters reported in [25].

B. Results and Analysis

We compare CG-HGN with several state-of-the-art 3-D instance segmentation methods on the four datasets, and report quantitative results in Tables I–IV. The corresponding qualitative visualizations are shown in Figs. 7 and 8. It is worth noting that CA-Cluster, SPFormer, and SphericalMask only predict instance segmentation results; therefore, their mIoU-S is not reported, and nonbuilding points in their visualizations are rendered in black.

TABLE I
BUILDING INSTANCE SEGMENTATION AND SEMANTIC SEGMENTATION RESULTS ON THE WHU3D DATASET

Method	Instance Metrics					Semantic Metrics	
	mAP	AP_{75}	AP_{50}	AP_{25}	mCov	mIoU-B	mIoU-S
CA-Cluster	20.70	20.45	27.78	29.06	69.52	46.77	-
B-Seg	31.91	31.97	51.32	68.07	44.08	72.09	50.78
SoftGroup	36.68	40.05	45.89	50.25	81.19	84.43	45.53
ISB-Net	28.52	27.18	40.19	48.82	67.75	90.38	60.75
SPFormer	30.26	29.65	37.30	46.42	64.80	79.13	-
Relation3D	32.19	30.77	42.02	54.45	60.29	95.15	-
SphericalMask	24.50	23.13	31.25	40.15	34.63	89.11	66.54
SuperCluster	45.39	45.76	56.09	60.75	77.38	**92.89**	77.72
CG-HGN (Ours)	**53.74**	**55.43**	**67.12**	**76.06**	**85.67**	92.70	**78.17**

TABLE II
BUILDING INSTANCE SEGMENTATION AND SEMANTIC SEGMENTATION RESULTS ON GUANGZHOU DATASET

Method	Instance Metrics					Sem. Metrics
	mAP	AP_{75}	AP_{50}	AP_{25}	mCov	mIoU-B
CA-Cluster	15.21	15.48	20.38	25.15	53.42	31.68
B-Seg	18.76	12.68	42.36	52.09	50.03	61.67
SoftGroup	28.48	31.38	48.30	55.79	73.30	63.71
ISB-Net	24.69	24.47	31.50	37.74	64.88	70.30
SPFormer	25.52	25.30	31.32	38.53	65.46	56.53
Relation3D	31.79	32.71	41.47	50.81	63.47	54.39
SphericalMask	23.29	22.53	27.60	31.82	42.36	53.71
SuperCluster	35.41	39.39	48.40	58.09	63.85	81.31
CG-HGN (Ours)	**47.05**	**52.05**	**64.38**	**72.03**	**74.12**	**83.82**

TABLE III
BUILDING INSTANCE SEGMENTATION AND SEMANTIC SEGMENTATION RESULTS ON THE STPLS3D DATASET

Method	Instance Metrics					Semantic Metrics	
	mAP	AP_{75}	AP_{50}	AP_{25}	mCov	mIoU	mIoU-S
CA-Cluster	34.18	35.55	40.98	44.87	83.46	48.47	-
B-Seg	59.02	62.12	70.61	82.90	75.44	88.66	73.29
SoftGroup	66.27	69.24	74.08	75.57	**88.34**	91.51	83.26
ISB-Net	55.19	56.35	68.00	77.38	79.29	**96.85**	**89.67**
SPFormer	47.37	47.31	56.96	66.39	72.00	92.94	-
Relation3D	57.90	57.61	71.15	82.97	75.60	95.81	-
SphericalMask	43.16	42.41	55.12	63.59	53.65	95.48	81.66
SuperCluster	68.67	74.34	79.46	81.31	88.05	95.95	86.98
CG-HGN (Ours)	**70.58**	**76.99**	**83.59**	**87.20**	87.43	96.31	86.77

TABLE IV
BUILDING INSTANCE SEGMENTATION AND SEMANTIC SEGMENTATION RESULTS ON THE URBANBIS DATASET

Method	Instance Metrics					Semantic Metrics	
	mAP	AP_{75}	AP_{50}	AP_{25}	mCov	mIoU	mIoU-S
CA-Cluster	18.93	18.72	22.30	24.29	73.23	69.51	-
B-Seg	34.30	32.72	57.25	69.49	23.56	58.55	44.23
SoftGroup	48.25	48.94	55.87	60.74	68.29	94.02	69.04
ISB-Net	41.14	41.27	47.43	52.35	64.65	95.59	**69.10**
SPFormer	42.57	41.78	49.22	56.72	58.30	93.61	-
Relation3D	43.39	41.90	50.62	60.05	53.33	95.12	-
SphericalMask	33.82	32.04	38.69	41.12	39.14	80.71	55.89
SuperCluster	54.07	55.42	62.22	65.08	65.84	96.19	64.54
CG-HGN (Ours)	**61.05**	**63.16**	**71.71**	**76.32**	**73.41**	**96.83**	66.07

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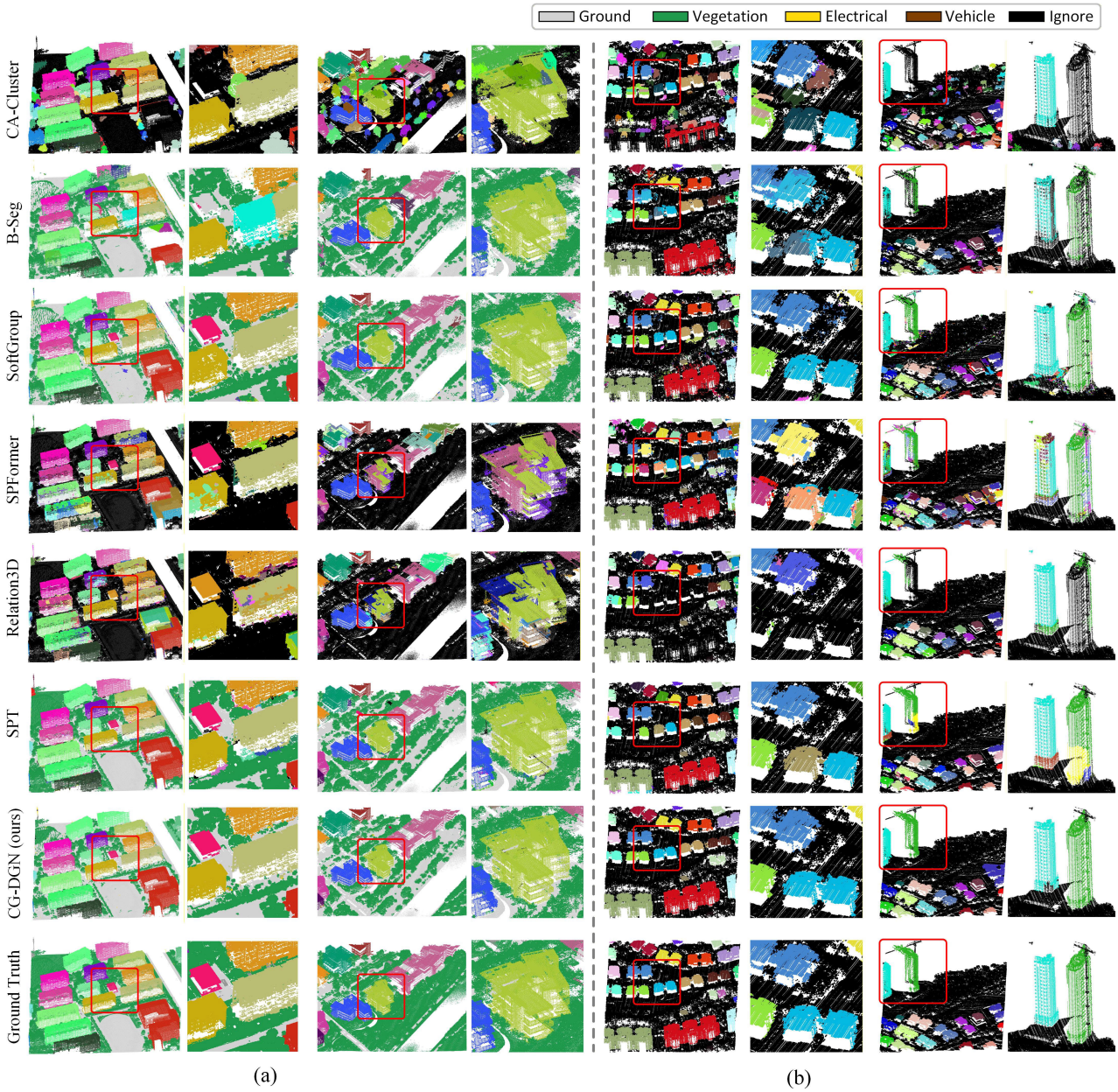

Fig. 7. Qualitative results on (a) WHU3D and (b) Guangzhou datasets. The scene size corresponds to a single batch item during training. Different colors indicate distinct building instances.

1) Whu3D: As shown in Table I, CG-HGN achieves a clear advantage on WHU3D in terms of instance segmentation metrics. The mAP reaches 53.74, outperforming the second-best method SuperCluster (45.39) by 8.35 percentage points. Under different IoU thresholds, the $AP_{75}/AP_{50}/AP_{25}$ of CG-HGN are 55.43/67.12/76.06, which improve over SuperCluster by 9.67, 11.03, and 15.31, respectively. The mCov increases from 77.38 to 85.67 (+ 8.29), indicating that CG-HGN provides better completeness and boundary coherence for building instances. At the semantic level, CG-HGN obtains a building mIoU-B of 92.70, which is essentially on par with SuperCluster (92.89). In the zoomed-in-view regions in Fig. 7(a), CG-HGN significantly alleviates boundary over-segmentation along building–ground interfaces. The hybrid graph network explicitly models edge–edge and edge–superpoint relations, enabling simultaneous improvements in high-threshold AP and overall coverage. Although CA-Cluster can identify most building entities, its heavy reliance on fixed-threshold geometric clustering limits its ability to distinguish buildings from surrounding vegetation, leading to a large number of FPs.

2) Guangzhou: In this scene, the severe point cloud incompleteness caused by occlusion poses higher requirements for the robustness of instance segmentation models. Although all methods show performance degradation on this dataset compared with WHU3D, CG-HGN still maintains a substantial lead. As reported in Table II, the mAP of CG-HGN is 47.05, improving over SuperCluster (35.41) by + 11.64. The $AP_{75}/AP_{50}/AP_{25}$ increase from 39.39/48.40/58.09 to 52.05/64.38/72.03, with absolute gains of + 12.66, + 15.98, and + 13.94, respectively. The mCov rises from 63.85 to

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74.12 (+ 10.27), showing that CG-HGN can preserve the spatial continuity of building structures even under sparse and heavily missing point distributions. On the semantic side, this dataset only provides building labels, so we only report mIoU-B. CG-HGN achieves a building mIoU-B of 83.82, higher than SuperCluster (81.31). As illustrated in the detailed views of Fig. 7(b), benefiting from the contour constraints and the topological consistency introduced by the edge graph, CG-HGN is capable of predicting relatively complete building boundaries even in regions with local points missing. In contrast, Transformer-based frameworks with querying (e.g., SPFormer and Relation3D) tend to attention localization during large-scale building recognition, leading to single buildings being split into multiple instances. Clustering-based methods (such as SoftGroup and B-Seg) mitigate over-segmentation to some extent by shifting points toward cluster centers.

3) STPLS3D: On this synthetic dataset, the regular geometric morphology and distinct boundaries of buildings contribute to an overall performance improvement across all methods compared with real-world scenes. CG-HGN continues to achieve the best results on primary instance segmentation metrics. As detailed in Table III, CG-HGN achieves an mAP of 70.58, representing a gain of + 1.91 over SuperCluster (68.67). The $AP_{75}/AP_{50}/AP_{25}$ scores improved from 74.34/79.46/81.31 to 76.99/83.59/87.2, corresponding to increases of + 2.65, + 4.13, and + 5.89. Regarding semantic accuracy, except for B-Seg, the mIoU-B for all methods exceeds 90, indicating that building semantic recognition on this dataset is approaching saturation. Specifically, the mIoU-B of CG-HGN is 96.31, slightly higher than that of SuperCluster at 95.95 (+ 0.36), while the nonbuilding mIoU-S is 86.77, which is comparable to SuperCluster's 86.98 (−0.21). In terms of the mCov metric, CG-HGN scores 87.43, slightly lower than SoftGroup (88.34) and SuperCluster (88.05), with differences remaining within 1 percentage point. This demonstrates that while maintaining comparable semantic accuracy and coverage, the proposed CG-HGN possesses a distinct advantage in recognizing building boundary structures.

4) UrbanBIS: On this dataset, CG-HGN achieves significantly better instance segmentation performance than the other methods. As reported in Table IV, mAP is improved from 54.07 (SuperCluster) to 61.05 (+ 6.98). The $AP_{75}/AP_{50}/AP_{25}$ increase from 55.42/62.22/65.08 to 63.16/71.71/76.32, with gains of + 7.74, + 9.49, and + 11.24, respectively. The mCov rises from 65.84 to 73.41 (+ 7.57). For semantics, CG-HGN achieves a mIoU-B of 96.83, higher than SuperCluster (96.19, + 0.64). The nonbuilding mIoU-S reaches 66.07, again outperforming SuperCluster (64.54, + 1.53), indicating consistent improvements in both building boundary recognition and background region delineation. As illustrated in Fig. 8(b), for ring-shaped buildings partially occluded by vegetation, SuperCluster produces multiple separate instances, whereas CG-HGN, benefiting from the topological priors provided by contours and the hybrid graph architecture, yields more complete and coherent building instances.

In summary, quantitative results across all four datasets reveal that CG-HGN achieves leading performance in mAP and AP metrics, with significantly superior mCov on WHU3D, Guangzhou, and UrbanBIS. The simultaneous rise in both high-threshold (AP_{75}) and low-threshold (AP_{25}) scores highlights enhanced instance discrimination alongside geometric integrity. The improvement in building mIoU-B confirms that contour features refine boundary representation without compromising background semantic generalization. Qualitatively, our method effectively mitigates over-segmentation, preserving the integrity of large-scale or ring-shaped buildings. Furthermore, the contour constraints and hybrid graph jointly ensure spatial coherence even under occlusion, directly contributing to the observed gains in mCov and high-precision AP.

TABLE V
COMPARISON OF TRAINING AND INFERENCE TIME ON THE WHU3D DATASET

Method	Train time	Epoch	Inference time (ms)
CA-Cluster	/	/	6839
B-Seg	9h 02m	500	$525 = 113 + 412$
SoftGroup	13h 58m	300	$846 = 121 + 725$
ISB-Net	8h 17m	300	$302 = 197 + 105$
SPFormer	9h 30m	1000	$604 = 65 + 539$
Relation3D	9h 28m	1000	$1192 = 49 + 1143$
SphericalMask	15h 54m	1000	$410 = 52 + 358$
SuperCluster	5h 41m	2000	$969 = 220 + 749$
CG-HGN (Ours)	8h 23m	2000	$516 = 461 + 55$

C. Training and Inference Efficiency

Table V presents a comparison of training and inference efficiency between CG-HGN and other instance segmentation models on the WHU3D dataset. The inference time is decomposed into the backbone feature extraction stage (the first number) and the subsequent instance segmentation stage (the second number). Note that the reported times exclude data preprocessing.

As observed from the table, in the training phase, although the hybrid graph structure introduces certain additional computational overhead, the total training time of CG-HGN remains at a moderate level among comparable methods, being significantly faster than complex frameworks such as SoftGroup and SphericalMask. In the inference phase, the total inference time of CG-HGN is 516 ms ($461 + 55$), which is significantly superior to the graph-cut-based SuperCluster 969 ms ($220 + 749$). This efficiency is primarily attributed to our adoption of binary connectivity prediction on the edge graph, which allows instance generation via a single connected component extraction. This approach is far more efficient than the optimization required by graph-cut algorithms on large-scale graphs. In contrast, while CA-Cluster is training-free, its contour-based geometric clustering is computationally intensive during inference; the average inference time of 6839 ms limits its practical applicability in large-scale urban scenarios.

D. Ablation Study

We conducted a series of ablation studies on the WHU3D and UrbanBIS datasets, as seen in Table VI. Reported metrics include mAP, AP_{50}, AP_{25}, and mCov. The "Best Model" row

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Fig. 8. Qualitative results on (a) STPLS3D and (b) UrbanBIS datasets. The scene size corresponds to a single batch item during training. Different colors indicate distinct building instances.

TABLE VI
ABLATION STUDY OF KEY COMPONENTS ON THE WHU3D AND URBANBIS DATASETS

Experiment	WHU3D				UrbanBIS			
	mAP	AP_{50}	AP_{25}	mCov	mAP	AP_{50}	AP_{25}	mCov
Best Model	53.74	67.12	76.06	85.67	61.05	71.71	76.32	73.41
(a) w/o Contour Feat.	−2.24	−2.02	−0.37	−0.84	−1.53	−0.91	−1.05	−2.09
(b) w/o Hybrid Graph	−6.03	−5.06	−8.17	−7.14	−6.58	−8.66	−10.35	−5.34
(b) w/o Cross-Graph Inter.	−4.85	−3.76	−2.71	−3.95	−3.60	−1.93	−3.55	−4.13
(c) MLP Fusion	−9.41	−9.21	−7.88	−5.94	−3.49	−5.55	−4.92	−5.20
(d) w/o Edge Pred	−6.34	−5.50	−5.26	−9.07	−7.17	−8.68	−7.84	−4.81
(e) Random Sampling	−1.49	−1.14	−0.15	−0.36	−0.25	−0.29	−0.35	−3.49
(e) Reduced Edges	−1.58	−2.06	−2.47	−3.79	−2.75	−2.53	−1.48	−1.14
(f) w/o Stage-wise	−5.16	−1.63	−0.54	−4.59	−5.26	−6.38	−5.76	−5.98

presents the absolute performance values of the full CG-HGN, while the remaining rows indicate the performance differentials relative to the best model.

1) Impact of Contour Features: When contour features are removed, the model exhibits a distinct performance decline on both datasets. On WHU3D, mAP and mCov decrease by

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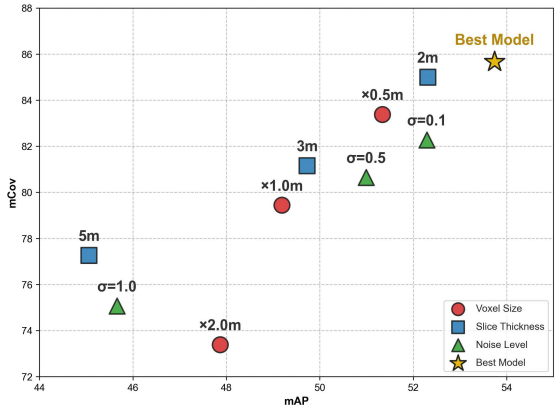

Fig. 9. Sensitivity and robustness analysis of contour features on WHU3D.

2.24 and 0.84, respectively; on UrbanBIS, mAP drops by 1.53 and mCov by 2.09. This indicates that contour features provide effective geometric shape priors, enabling the network to better capture building stratification, thereby improving instance segmentation accuracy at building boundaries. Without these features, the model's capability to recognize complex boundaries is significantly weakened.

We also investigated the stability of the contour features by evaluating the model's sensitivity to extraction contour hyperparameters (voxel sizes and slice thicknesses) and its robustness against injected Gaussian noise (σ) on the WHU3D validation set, as illustrated in Fig. 9.

Regarding contour hyperparameter sensitivity, the model remains stable within a moderate range but degrades under excessively coarse settings. Specifically, a 5-m slice thickness or a $\times$2.0-m voxel size over-smooths structural details, dropping mAP to 45.06 and 47.87, respectively. In contrast, finer settings (e.g., a 2-m slice) effectively preserve boundary cues and yield results highly competitive with the base model.

Furthermore, CG-HGN exhibits strong robustness to contour extraction artifacts. When simulated with Gaussian noise, the model maintains an mAP $>$ 50 under mild to moderate noise ($\sigma = 0.1, 0.5$) and avoids catastrophic failure even under severe noise ($\sigma = 1.0$). This resilience is attributed to the dynamic gating mechanism, which adaptively downweights irregular contour descriptors in favor of stable local geometric features.

2) Impact of Hybrid Graph Structure: We ablate two core components of the hybrid graph architecture: the edge graph itself ("w/o hybrid graph") and the bidirectional feature exchange ("w/o cross-graph inter"). Retaining only the superpoint graph causes the most severe performance drop, with mAP and mCov falling by 6.03 and 7.14 on WHU3D, and mAP dropping by 6.58 on UrbanBIS. This confirms the edge graph's critical role in supplementing topological dependencies. Furthermore, when the edge graph is kept, but the cross-graph feature interaction is disabled, the performance still declines substantially. Specifically, mAP and mCov drop by 4.85 and 3.95 on WHU3D and by 3.60 and 4.13 on UrbanBIS. Without cross-graph interaction, the superpoint graph lacks the boundary-aware cues necessary to restrict cross-instance message passing, while the edge graph is deprived of updated semantic contexts. This confirms that dynamic relational learning is essential for mitigating over-segmentation and achieving high instance coverage.

3) Impact of Feature Fusion Mechanism: To verify the necessity of our proposed dynamic gated feature fusion mechanism, we established a baseline experiment using simple concatenation followed by an MLP. Compared with simple concatenation, incorporating dynamic gating improves mAP on WHU3D and UrbanBIS by 9.41 and 3.49 points, respectively, with corresponding mCov increases of 5.94 and 5.20. This suggests that simple concatenation fails to fully exploit the contribution of contour features. In contrast, the gating mechanism explicitly regulates the relative weights of local geometry and contour features, enabling the network to adaptively control its reliance on contour information, thus yielding superior instance segmentation performance.

The spatial and statistical distributions of the gating parameter g_s on the UrbanBIS and Guangzhou datasets demonstrate the effectiveness of this mechanism. Following (1), g_s explicitly controls the weight of the contour features F_s^c fused into the local geometric features F_s^g. Fig. 10 reveals that this spatial distribution is highly context-aware.

1) *Building Tops (Low g_s, Blue Areas):* Given the complete point coverage and stable local geometry at building tops, the network assigns lower g_s values, relying primarily on local features rather than global contours.
2) *Building Facades (Gradient to High g_s):* As occlusions and missing points increase closer to the ground, the network adaptively raises g_s (shifting toward yellow and red) to leverage contour priors for structural integrity.
3) *Ground and Intersections (High g_s, Red Areas):* To avoid over-segmentation at ambiguous building–ground boundaries, the model strongly utilizes contour features as explicit structural constraints, resulting in high g_s values.

Furthermore, the histograms display a broad distribution of g_s. This statistical evidence proves that the dynamic gating mechanism successfully achieves adaptive feature selection based on the specific geometric context of each superpoint, rather than collapsing to a static weight during training.

4) Impact of Edge Prediction Mechanism: We replaced the graph connectivity segmentation in our network with graph-cut and changed the edge graph output from binary connectivity prediction to edge affinity prediction (as used in SuperCluster). Results show a decline in performance: on WHU3D, mAP and mCov decrease by 6.34 and 9.07; on UrbanBIS, mAP drops by 7.17 and mCov by 4.81. This indicates that within a hybrid graph network, direct binary edge connectivity prediction is more suitable for urban building instance segmentation tasks than continuous affinity modeling, as it effectively reduces boundary ambiguity and erroneous connections.

5) Impact of Edge Quantity and Sampling Strategy: To analyze the influence of edge quantity of the edge graph and sampling methods, we tested random edge sampling and reduced edge quantities. Using random sampling causes mAP drops of 1.49 on WHU3D and 0.25 on UrbanBIS, suggesting that random sampling disrupts key topological structural information, destabilizing the edge relationships learned by the

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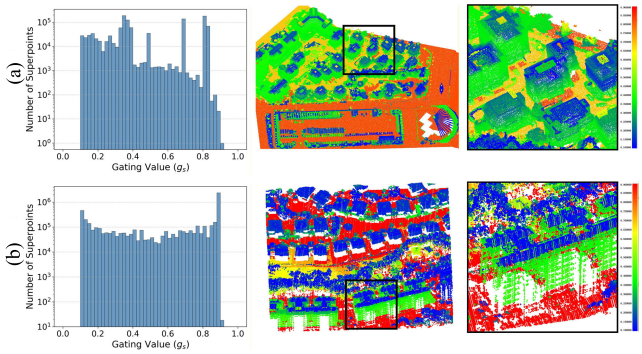

Fig. 10. Spatial and statistical distributions of the dynamic gating parameter g_s on (a) UrbanBIS and (b) Guangzhou datasets. (Left) Histograms demonstrate the broad numerical distribution of g_s. (Right) Heatmaps illustrate its spatial adaptivity, where red represents a higher fusion of morphological features and blue represents a lower fusion.

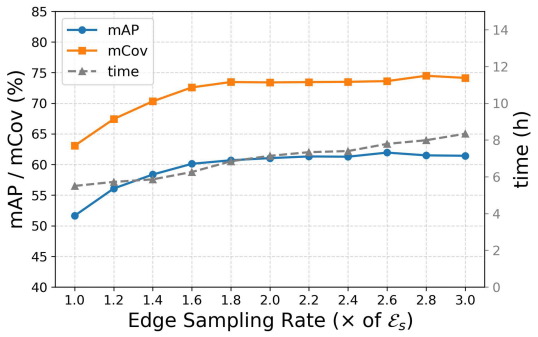

Fig. 11. Sensitivity analysis of model performance and training time under different edge sampling rates (1.0×–3.0× scale of superpoint graph edges) on the UrbanBIS dataset.

network. Furthermore, when reducing the number of edges in the edge graph to 1.5× the number of superpoint graph edges, mAP on WHU3D and UrbanBIS decreases by 1.58 and 2.75, respectively. This implies that appropriate redundancy in edge connections is crucial for maintaining global topological integrity.

Building on this, we conducted systematic testing on the UrbanBIS dataset with varying edge sampling rates, gradually increasing the edge graph scale from 1.0× to 3.0× the superpoint graph edge set. The results are illustrated in Fig. 11. As the edge sampling rate increases, both mAP and mCov metrics show a steady upward trend, indicating that increasing edge quantity within a certain range helps to enhance graph structure representation and discriminative performance. Meanwhile, the corresponding training time increases from approximately 5.5–8.3 h, reflecting a nearly linear coupling between edge quantity and computational cost. Therefore, selecting a reasonable edge sampling rate is essential for balancing performance and efficiency: over-sampling benefits performance but significantly increases training overhead, while under-sampling weakens structural information representation.

6) Impact of Stagewise Training Strategy: Finally, we evaluated the impact of removing the stagewise training strategy. Results show that this strategy is critical for both model convergence and final accuracy. On WHU3D, mAP and mCov

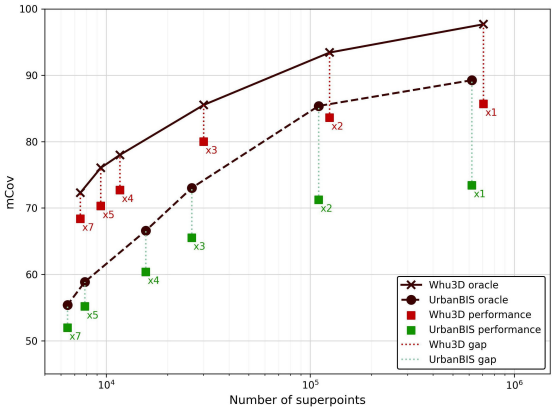

Fig. 12. Quantitative evaluation of model sensitivity to superpoint granularity. The "Oracle" curves represent the theoretical upper bound of mCov given a specific superpoint partition, while "Performance" curves denote the actual model results. The regularization strength of cut-pursuit increases from right to left (×1–×7), resulting in fewer, coarser superpoints.

decrease by 5.16 and 4.59; on UrbanBIS, they drop by 5.26 and 5.98. This suggests that learning stable semantic representations on the superpoint graph first, followed by introducing edge graph relational constraints for joint optimization, facilitates better convergence and accuracy in complex tasks. Directly training the superpoint and edge graphs simultaneously makes it difficult for the model to fully leverage the information from the hybrid graph structure, resulting in a significant drop in instance segmentation accuracy.

E. Sensitivity to Superpoint Partition

In our framework, the final point-level building instance labels are directly inherited from their parent superpoints. Consequently, the model's overall performance is intrinsically linked to the granularity and quality of the initial superpoint partition. If a superpoint is under-segmented (e.g., spanning across an instance boundary or merging a building with the surrounding ground), all raw points within it are forced to share the same label.

To quantitatively evaluate this influence, we conducted a sensitivity analysis by scaling the regularization parameter in the cut-pursuit algorithm from ×1 to ×7 of our default baseline value used in the SPT64 backbone, thereby generating superpoints of different numbers. We evaluated these partitions on both the WHU3D and UrbanBIS datasets.

As shown in Fig. 12, we plot the performance of the "oracle" model, which associates each superpoint with its most frequent GT label. This model acts as a theoretical upper bound on the achievable segmentation performance with a given partition. As the regularization strength increases (moving left on the x-axis, resulting in fewer and coarser superpoints), the oracle mCov decreases significantly. This implies that excessively coarse superpoints adhere poorly to fine-grained boundaries between instances, thereby establishing a fundamental limit on the actual model performance.

We also illustrate the qualitative effects for different partitions of varying coarseness in Fig. 13. At the optimal scale (×1), building boundaries are well-preserved, providing a reliable foundation for graph-based reasoning. However, using

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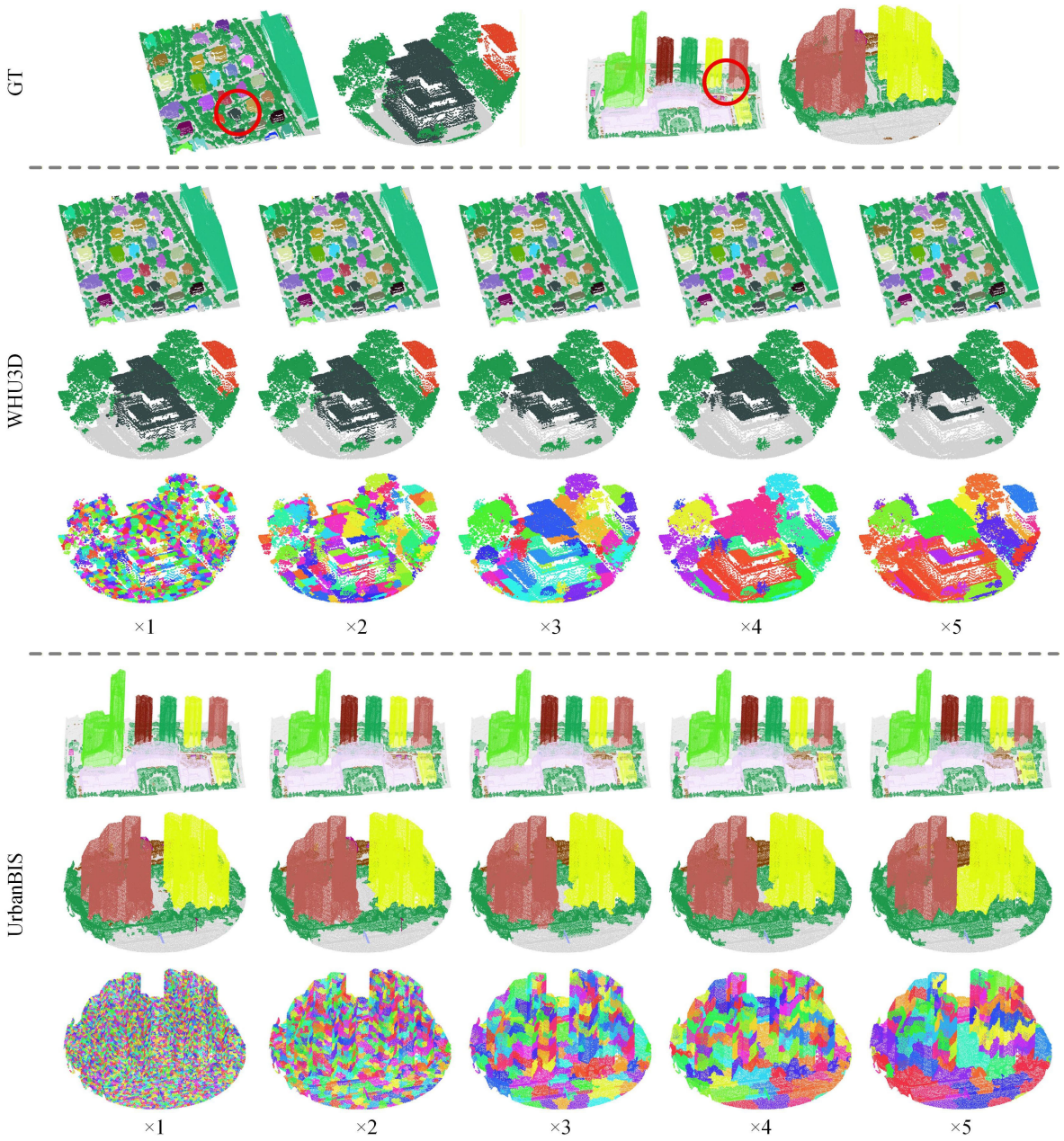

Fig. 13. Qualitative effects of varying superpoint partitioning scales on the WHU3D and UrbanBIS datasets. As the regularization parameter increases (from ×1 to ×5 the baseline value), coarse superpoints tend to merge adjacent buildings or blur building–ground boundaries, which limits the instance-level resolution of the final predictions.

substantially coarser superpoints (e.g., ×5) leads to a severe performance drop, as adjacent buildings or building–ground interfaces are erroneously merged within a single massive superpoint. Although the performance of our model at the finest scale (×1) remains below the oracle—suggesting that the partition does not strongly limit its performance at this level—the drastic degradation at coarser scales demonstrates that an excessively coarse partition eventually becomes the primary bottleneck, fundamentally restricting the resolution of the final instance outputs.

F. Generalization Ability

In real-world scenarios, many cities lack complete instance-level annotations for buildings, and manual annotation of 3-D scenes is prohibitively expensive and difficult to acquire at scale. To address this, we evaluated the generalization ability of different methods through cross-dataset training and testing configurations: training the model on one scene and testing it directly on others.

Fig. 14 presents quantitative comparisons under three cross-dataset configurations. Across different training–testing combinations, CG-HGN generally outperforms the SuperCluster method in terms of AP and mCov metrics. Although the absolute performance of all methods drops significantly in the cross-domain setting compared with intradataset results, CG-HGN maintains better instance consistency and boundary precision. This indicates that our model possesses strong structural transferability even in unseen scenes. The advantage

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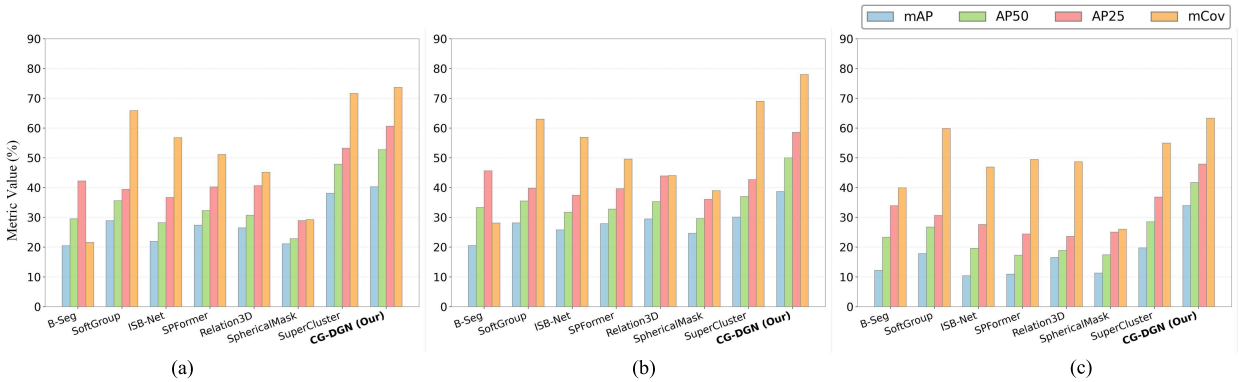

Fig. 14. Quantitative results of cross-dataset generalization for building instance segmentation. (a) Train: UrbanBIS and test: WHU3D. (b) Train: STPLS3D and test: WHU3D. (c) Train: WHU3D and test: Guangzhou.

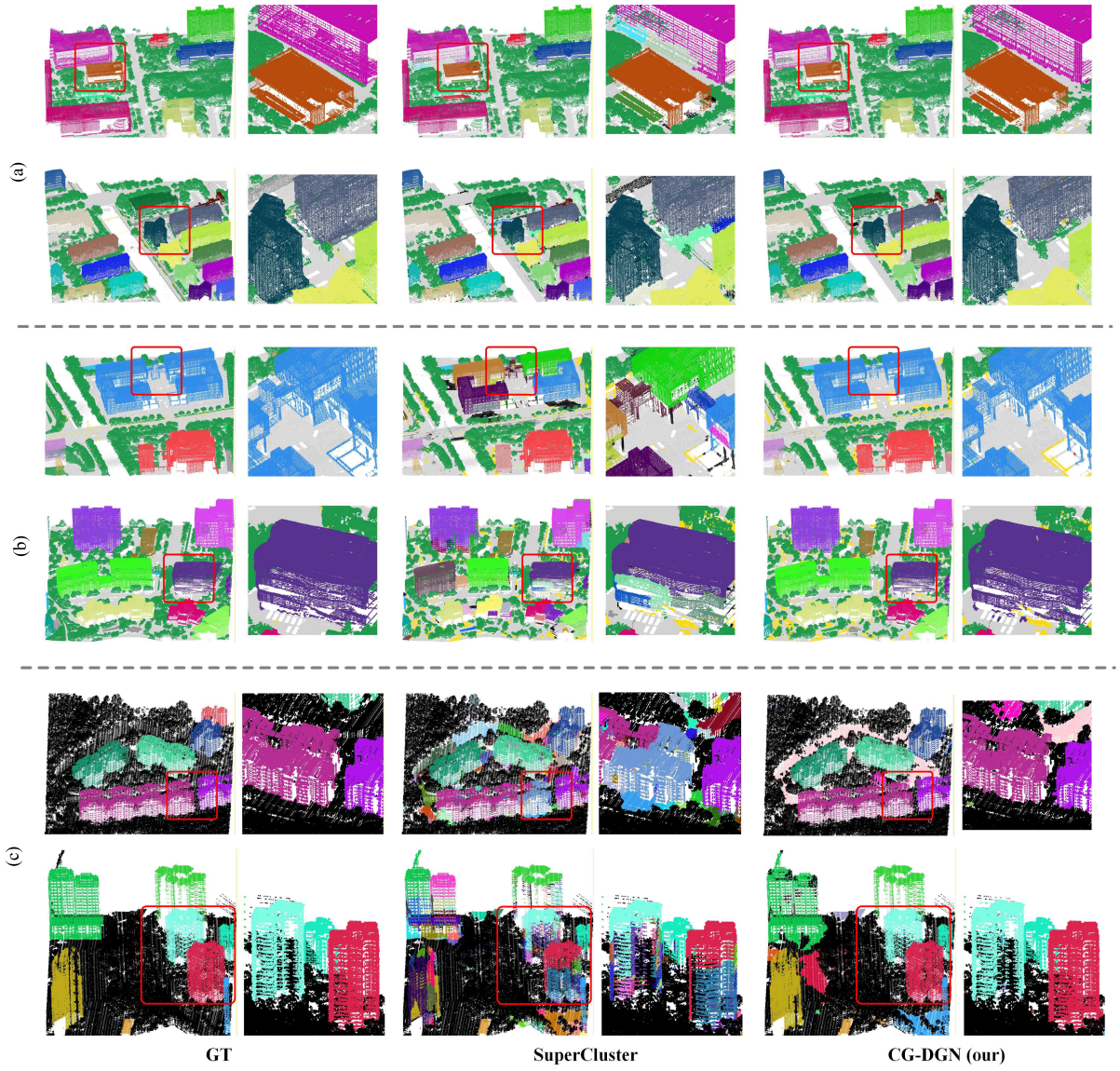

Fig. 15. Visual comparison of cross-dataset generalization for building instance segmentation. (a) Train: UrbanBIS and test: WHU3D. (b) Train: STPLS3D and test: WHU3D. (c) Train: WHU3D and test: Guangzhou.

is particularly evident in the mCov metric, suggesting that even when building morphology and point density vary, the model maintains high target recall and overall contour coverage.

Fig. 15 further provides visual comparisons between CG-HGN and SuperCluster under three cross-domain configurations. Overall, CG-HGN exhibits significantly less over-segmentation than SuperCluster. In the zoomed-in-view

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regions marked by red boxes: in Configurations 1 and 2, CG-HGN preserves better instance coherence on building facade areas; in Configuration 3, where Guangzhou scene suffers from severe occlusion and point cloud incompleteness—with many buildings distributed in mountainous terrain covered by dense vegetation—the difference from the training data is substantial. While performance for all methods degrades, CG-HGN still produces more accurate and coherent segmentation results than SuperCluster.

V. Conclusion

In this article, we have presented CG-HGN, an airborne LiDAR building instance segmentation method that integrates contour structure priors with hybrid graph collaborative reasoning. Compared with existing methods that rely solely on local geometry or point neighborhood relationships, CG-HGN explicitly models the contour morphology of buildings and the topological connections between superpoints, thereby achieving a stronger structural understanding. To effectively fuse heterogeneous features, we adopted a dynamic gating mechanism to adaptively integrate local geometric features with contour features. Furthermore, we constructed a hybrid graph network comprising a superpoint graph and an edge graph, enabling the collaborative optimization of semantic discrimination and instance boundary prediction within a unified framework.

Experiments on four datasets demonstrate that CG-HGN outperforms existing methods on key instance segmentation metrics such as mAP and AP_{75}. It also achieves significant advantages in mCov across most scenes, exhibiting superior coherence and robustness, particularly in real-world urban scenarios characterized by adjoining buildings or missing points. Leveraging binary connectivity prediction on the edge graph, our method generates instance results via a single-pass connected component extraction during inference, balancing high accuracy with inference efficiency.

Overall, CG-HGN offers a novel perspective on utilizing morphological priors. However, limitations remain that define our future work. First, the hybrid graph modeling incurs higher computational costs than point-only networks, necessitating the design of lighter weight modules. Second, the boundary resolution of our pure LiDAR-based model is inherently constrained by the initial superpoint partition and the absence of nongeometric priors. Consequently, in extremely dense urban areas, narrow gaps may suffer from superpoint under-segmentation, and physically connected buildings are naturally merged into continuous geometric envelopes. Separating such complex instances motivates future research into adaptive subdivision mechanisms and multimodal fusion (e.g., incorporating imagery or cadastral maps). Finally, reliance on aerial contour priors limits applicability to MLS data; future iterations will integrate facade features to enhance multisensor adaptability and extend the framework to general 3-D instance tasks.

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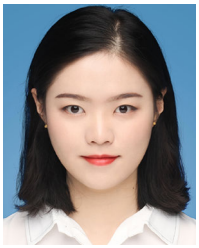

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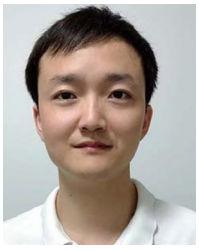

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